

Getting Started with Python

A friendly beginner's guide to the basics

1. What Is Python?

Python is a general-purpose programming language known for its clean, readable syntax. It's used for web development, data analysis, automation, scientific computing, and much more. It's often recommended as a first language because the code reads almost like plain English.

2. Your First Program

Every Python tutorial starts here. Save the following in a file called **hello.py** and run it with **python hello.py**:

```
print("Hello, World!")
```

The **print()** function displays text on the screen. That's it — no extra setup required.

3. Variables and Data Types

Variables store data so you can use it later. Python figures out the type automatically — you don't have to declare it.

```
name = "Ada"           # string (text)
age = 30                # integer (whole number)
height = 1.65           # float (decimal number)
is_coder = True         # boolean (True/False)
```

- **str** — text, written in quotes: "hello"
- **int** — whole numbers: 7, -3, 100
- **float** — decimal numbers: 3.14, 0.5
- **bool** — True or False

4. Basic Operations

```
x = 10
y = 3
```

```
print(x + y)  # 13  addition
print(x - y)  # 7   subtraction
print(x * y)  # 30  multiplication
print(x / y)  # 3.33... division
print(x % y)  # 1   remainder
```

```
print(x ** 2) # 100 exponent
```

5. Lists

A list holds an ordered collection of items, and you can change it after creating it.

```
fruits = ["apple", "banana", "cherry"]

print(fruits[0])      # apple (indexing starts at 0)
fruits.append("mango") # add an item
print(len(fruits))    # 4 (number of items)
```

6. Conditionals

Use **if**, **elif**, and **else** to make decisions in your code. Indentation (usually 4 spaces) defines what belongs inside each block — this is required in Python, not optional.

```
age = 18

if age >= 18:
    print("You can vote")
elif age >= 16:
    print("Almost there")
else:
    print("Not yet")
```

7. Loops

A **for** loop repeats code for each item in a sequence. A **while** loop repeats as long as a condition stays true.

```
for fruit in fruits:
    print(fruit)

count = 0
while count < 3:
    print("Counting:", count)
    count += 1
```

8. Functions

Functions let you package up reusable blocks of code.

```
def greet(name):
    return f"Hello, {name}!"

message = greet("Ada")
print(message)    # Hello, Ada!
```

9. A Few Beginner-Friendly Habits

- Run your code often — after every small change, not just at the end.
- Read error messages closely; they usually point to the exact line and issue.
- Use descriptive variable names (total_price, not tp).
- Break big problems into small functions.
- Don't memorize everything — look things up. Every developer does.

10. Where to Go Next

- Practice with small projects: a calculator, a to-do list, a number-guessing game.
- Learn about dictionaries and tuples (other core data structures).
- Explore libraries for what interests you: requests (web), pandas (data), pygame (games).
- The official Python tutorial at docs.python.org is a solid, free next step.

You now know enough to write real, working Python programs. The best way to learn from here is to build something small and keep going.